Debarshi Chakraborty

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Education

- **Ramakrishna Mission Vivekananda Educational and Research Institute**
  Master of Science - Big Data Analytics — *July 2024 - Present*
- **Ramakrishna Mission Vivekananda Centenary College**
  Bachelor of Science - Mathematics Honors — *September 2021 - July 2024*

Experience

- **AI/ML Intern** — January 2026 - Present
  Rezolve AI — *Kolkata, West Bengal*
  - Engineered a production **Agent-to-Agent (A2A)** multi-agent system end-to-end — orchestrator design, sub-agent prompt specialization, **MCP** tool server deployment, dynamic skill routing, and **Redis**-backed cross-agent context sharing — powering a live retail conversational AI platform.
  - Built a **long-term memory** layer using a dedicated **pgvector** semantic retrieval agent, injecting past user preferences and session context into the A2A orchestration flow, reducing redundant inputs by **45%** and improving cross-session recommendation relevance.
  - Refactored the retail **LLM prompt stack** into a **3-layer architecture** (persona, tonality, task) managed in **Langfuse**, standardizing behavior across **6+ sub-agents** and reducing prompt-related production incidents by **~40%**.
  - Benchmarked an internal **120B model** against **Claude Sonnet 4.5** for A2A tool calling — achieving **100% routing accuracy** and **20–25% latency reduction**, directly informing production model selection.
  - Implemented **multimodal input routing** for image-only and image+text queries and an **SSE-streamed context-aware chip generation** layer — improving session engagement depth by **~30%**.
- **Summer Research Intern** — May 2025 - July 2025
  Indian Institute of Technology, Guwahati — *Guwahati, India*
  - Collaborated with **PhD researchers** to design **graph algorithms** for the **Popular Matching Problem** in subcubic graphs and bounded-degree graph families.
  - Developed and analyzed **combinatorial** and **reduction-based algorithms** to determine the existence of popular matchings under structural constraints; implemented prototypes in **Python**.

Projects

- **Real-Time (T+0) Trade Settlement System for the US Market**:
  - Architected a 4-node permissioned blockchain network using **Hyperledger Besu** with **IBFT-2 consensus** for deterministic, atomic delivery-versus-payment (DvP) settlement of tokenized securities and cash.
  - Integrated **LangGraph-based state machine agents** (Llama 3.2) for pre-trade compliance validation and real-time liquidity checks; applied **SHAP and LIME** for explainable anomaly detection.
  - Deployed **Random Forest and XGBoost** fraud detection engines, achieving **91.7% reduction** in counterparty risk. Published at **IEEE WCONF 2025**.
  - **Technologies:** Hyperledger Besu, Smart Contracts, Python, LangGraph, LLMs, XGBoost
- **NET-LINK: Self-Healing Drone Swarms**:
  - Built a custom **cooperative multi-agent** environment (**Dec-POMDP**) where 5 autonomous drones maintain a dynamic **communication relay** under energy, collision, and boundary constraints.
  - Benchmarked **MAPPO**, **MADDPG**, and **QMIX** under the **CTDE** paradigm; MAPPO achieved the best coordination with **45% success rate** and near-zero collisions.
  - **Technologies:** PyTorch, MAPPO, MADDPG, QMIX, MuJoCo, Custom Gym Environment
- **Distributed Inference for Large Language Models**:
  - Deployed a **distributed inference** cluster for **DeepSeek R1 Distill** and **LLaMA 3.2B Instruct** using **model and tensor parallelism**, achieving a **2–4× increase** in tokens/sec throughput.
  - **Technologies:** Python, C++, CUDA, Distributed Computing
- **Custom Deep Learning Document Summarization and QA System**:
  - Built a **GRU-based Seq2Seq** model with **attention mechanism** and **beam search** for document summarization, and fine-tuned **DistilBERT** on **SQuAD 2.0** for **extractive QA** achieving **72.85% F1**, outperforming classical RNN baselines.
  - **Technologies:** Python, PyTorch, Hugging Face Transformers, NLP
- **Neural US Equity Option Pricing**:

- Identified and resolved **Martingale Leakage** in **LSTM** regression via **stationary feature engineering** and a hybrid **Black-Scholes** residual learning approach.
- Pivoted to **directional classification**, achieving **60.39% test accuracy** vs. 50% random walk baseline with a stable 1.52% generalization gap confirming extractable **momentum alpha** in high-frequency markets.
- **Technologies:** Python, PyTorch, LSTM, Time Series Analysis, Quantitative Finance

Skills Summary

- **Languages**: Python, R, Java, C++, SQL
- **AI/ML**: PyTorch, Scikit-learn, NLTK, LangGraph, LangChain, OpenAI Gym, GNNs
- **MLOps & Infra**: Docker, Langfuse, Qdrant, Weaviate, FastAPI, Streamlit, Azure OpenAI
- **Data**: NumPy, Pandas, PySpark, Dask, Polars, Matplotlib
- **Tools**: Git, GitHub, VS Code, Jupyter, Neo4j

Achievements

- **IEEE WCONF 2025 Publication** *2025*
  Published *"Real Time Trade Settlement for the US Market"* at the 3rd World Conference on Communication & Computing. DOI: 10.1109/WCONF64849.2025.11233239

- **Cleared IIT JAM 2024** *February 2024*
  Successfully cleared the IIT JAM 2024 examination in Mathematics, securing eligibility for master's programs at IITs.

Volunteer Experience

- **Event Lead: CodeCraft — Perceptron 26, RKMVERI Annual Tech Fest** *January 2026*
  Spearheaded the flagship Hackathon "CodeCraft", managing logistics, problem statements, and participant coordination for 300+ attendees.